PHILIP UMEADI

philipumeadi@icloud.com | https://mindphil.github.io/

EDUCATION

Syracuse University Syracuse, NY

College of Arts & Sciences August 2023 – May 2026

Bachelor of Science in Applied Mathematics, Minor in Economics GPA: 3.16

Relevant Coursework: Real Analysis, Probability Theory, Stochastic Processes, Numerical Methods, Linear Algebra, ODEs/PDEs

RELEVANT WORK EXPERIENCE

Saint Joseph's Preparatory School, Math Teacher, Philadelphia, PA August 2026 – Present

- Teaching algebra and geometry courses under tenured math faculty

Verisk, Analyst Intern, Jersey City, NJ June 2025 – August 2025

- Collaborated with senior developers to engineer a scalable Python framework to complete a vendor record migration of 27,000+ legacy documents to a new database ahead of contract renegotiations.
- Built a Python automation suite that replaced hand-editing recurring vendor correspondence

Extraordinary Re, Research Intern, New York, NY May 2023 – August 2023

- Researched how a consortium model could attract institutional participants to a marketplace for trading insurance contracts and
- Recruited a former GE Private Equity VP to spearhead operations

RESEARCH & PROJECTS

Document Filing Pipeline, Richard Health Systems June 2026

- Built an offline Python pipeline that splits, classifies, and files scanned employee onboarding PDFs into the correct records folders by name, replacing the manual sort of 15–20 stacked documents per new hire
- Added a confidence gate that auto-files only high-certainty matches and routes the rest to a review queue, keeping accuracy on messy duplex scans while all sensitive employee records stay on the local machine

Mathematical Foundations of Machine Learning, Syracuse University January 2026 – May 2026

- Supervised (by Justin Ko, PhD) reading course through three standard ML texts (Deisenroth; James; Jurafsky & Martin), implementing selected methods in Python.
- Built a nonlinear regression model in Python (polynomial–sinusoidal basis, MLE) to forecast a year of natural gas prices, tuning model complexity on held-out data so it captured the seasonal pattern.
- Built a character-level n-gram model in Python to play Hangman, interpolating unigram–trigram with perplexity-tuned weights to average 8.2 wrong guesses per word, beating a 14-guess frequency baseline

Binary Fission as a Homogeneous Branching Process, Independent Project March 2026

- Modeled antibiotic-driven bacterial colony extinction as a homogeneous branching process, deriving a closed-form threshold for the minimum drug concentration that guarantees extinction
- Compared two real antibiotics' theoretical efficacy using published pharmacodynamic parameters (Bogdanov et al.), then built a Python/Manim simulation animating the stochastic process
- Modeled the extinction of bacterial colonies under antibiotic treatment as a homogeneous branching process, deriving a closed-form threshold for the minimum drug concentration that guarantees extinction

Linear Algebra and Category Theory, Syracuse University January 2025 – April 2025

- Studied how large linear systems break into irreducible building blocks (the Kronecker problem).

Combinatorial Game Theory, Syracuse University August 2024 – November 2024

- Investigated a network pursuit game (an n-cop generalization of the Shannon Switching Game, due to K. Wood) and conjectured that the resources needed grow linearly with network size
- Presented findings at a seminar (Jan. 2026) and spurred programmatic and combinatorial attempts to prove a linear upper bound

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Tools: Git, Docker, REST/LLM APIs, Bloomberg Terminal, Excel/VBA, BeautifulSoup

LEADERSHIP & ACTIVITIES

Syracuse University Division 1 Rowing August 2023 – June 2026

- 20+ hours a week of training while carrying a full applied-math course load.
- Finished with an undefeated regular season and a 4th place finish at the national championship regatta

Manhattan College FED Challenge & Investment Club September 2022 – May 2023

- Built econometric models forecasting CPI, unemployment, and GDP
- Conducted DCF and comparable company analysis on equity positions for student-managed portfolio